

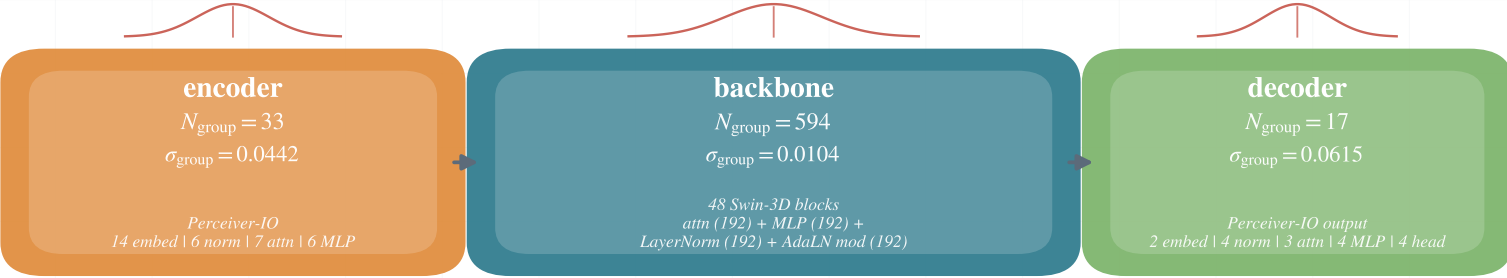
Weight perturbation across four AI weather models



Aurora

Swin-Transformer 3D U-Net

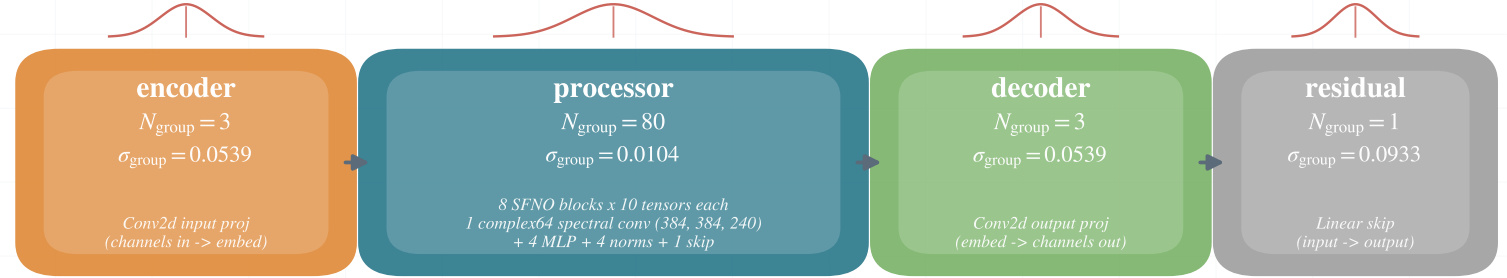
$N_{\text{total}} = 644$



SFNO

Spherical Fourier Neural Operator

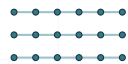
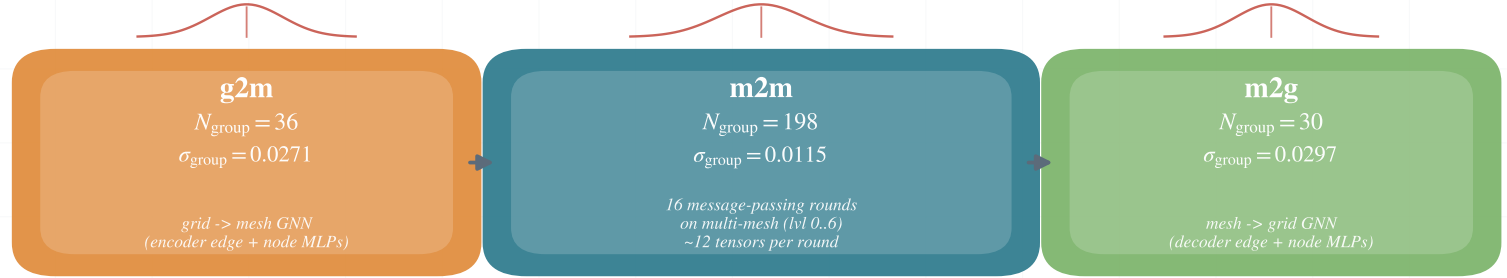
$N_{\text{total}} = 87$



GraphCast

Multi-Mesh Graph Neural Network

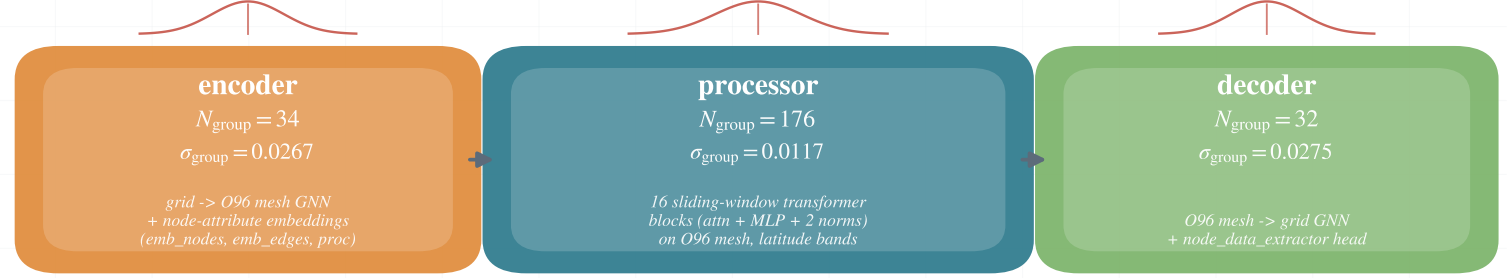
$N_{\text{total}} = 264$



AIFS

GNN + latitude-band transformer

$N_{\text{total}} = 242$



encoder

processor

decoder

aux / residual

 = Gaussian noise injection on every learnable tensor in the group